

# SMART ENVIRONMENT MONITORING SYSTEM

## Abstract

This project presents a Smart Environment Monitoring System designed to measure and visualize environmental parameters such as temperature, humidity, and gas levels in real time. The system uses ESP32 as an IoT sensing node connected to DHT11 and MQ-2 sensors for data acquisition.

The ESP32 publishes sensor readings using the MQTT protocol. The data is received by an MQTT subscriber application, stored in a MySQL database, optionally uploaded to ThingSpeak cloud, and displayed on a Flask-based web dashboard created using HTML. The project demonstrates IoT communication, cloud integration, local database storage, and real-time visualization.

## Aim

The aim of this project is to design and implement a smart environment monitoring system capable of:

- Collecting environmental parameters in real time
- Sending data wirelessly over MQTT
- Storing sensor readings in a MySQL database, ThingSpeak and web-based dashboard.

## Objectives

1. To interface DHT11 and MQ-2 sensors with ESP32 microcontroller.
2. To implement MQTT protocol for wireless data communication.
3. To demonstrate complete IoT architecture from sensing to visualization.

## System Overview

The ESP32 reads temperature, humidity, and gas values from DHT11 and MQ-2 sensors. The ESP32 acts as an **MQTT publisher** and sends data to the MQTT broker. A Python-based MQTT subscriber receives messages, stores the data in MySQL database, uploads data to ThingSpeak, and sends values to Flask-based dashboard. Users can monitor environmental readings live via dashboard and cloud.

## Team Roles

### 94104 – TANISHA GULAWANI

- Publisher code development on Arduino ESP32
- MQTT publish and subscribe testing
- ESP32 Wi-Fi & broker connection
- Database creation and query execution

### **94101 – ANKITA MOLEKAR**

- MQTT subscriber code (Python – VS Code)
- MySQL database connection and table insertion
- MQTT message receiving and parsing

### **94105 – OM PATIL**

- Dashboard creation using HTML and CSS
- Flask dashboard backend code (dashboard.py)
- Connecting dashboard with MySQL database and project

### **Conclusion**

The Smart Environment Monitoring System successfully demonstrates IoT-based sensing, wireless communication, cloud integration, local database storage, and real-time dashboard visualization. The use of MQTT protocol makes communication lightweight and efficient. The project is scalable and can be extended for smart homes, industries, and environmental monitoring applications.

### **References**

1. ESP32 Technical Documentation – Espressif Systems
2. MQTT Protocol Official Documentation – [mqtt.org](https://mqtt.org)
3. ThingSpeak Cloud API Documentation